Measure first.
Diagnose with evidence.

Link health on a disrupted or constrained network

Peer presence + round-trip time + reliable-delivery counters

Use existing traffic. Keep measurement separate from diagnosis.

Three questions need three measurements

Question	Measurement	What it tells us
Can I reach this peer?	`RouterHealth` presence	Current reachability state
How long is the round trip?	Passive heartbeat RTT	Delay, after subtracting the peer's reporting wait
Is reliable delivery under pressure?	Per-peer DDS protocol counters	Repairs, backlog, and receiver pressure

RTT = round-trip time. DDS = Data Distribution Service.

> **Takeaway:** reachability is not delivery quality, and one counter is not a root cause.

Measure each peer, not just the router total

Local endpoint	Remote peer	Useful evidence
Writer	Matched reader	Requested repairs and unacknowledged data
Reader	Matched writer	Missing sequences, duplicates, and rejections

- Poll cumulative counters and calculate interval differences locally.
- Keep separate baselines for each endpoint/peer association.
- Restart, rematch, or counter reset starts a new baseline.

Takeaway: do not report replay or a new endpoint association as sudden link degradation.

Separate network symptoms from local pressure

Signal	Useful interpretation	Not proof of
Repair requests and retransmissions	Data needed recovery	RF loss alone; late joins and replay also matter
Unacknowledged backlog	Sender is waiting for progress	Network congestion alone
Samples waiting behind a sequence gap	In-order delivery is blocked	The source of the missing sample
Rejected samples / full resources	Receiver or sender pressure	A failed radio link

NACK = request to repair missing data.

> **Takeaway:** correlate counters with packet capture, host load, and controlled tests.

RTT reuses existing heartbeats

1. A sends a scheduled `RouterHealth` heartbeat.
2. B records receipt and acknowledges it in B's next scheduled heartbeat.
3. A subtracts B's reported waiting time from A's elapsed round trip.

```
RTT = A's elapsed time - B's receipt-to-report wait
Example: 725 ms - 680 ms = 45 ms
```

- No dedicated `RouterLinkProbe` stream in the current design.
- No subtraction of wall-clock timestamps across machines.
- Dispatch, queueing, and the return path still contribute.

> **Takeaway:** this is a system round trip, not pure propagation delay.

Missing or stale evidence is not zero latency

Displayed fact	Meaning
Presence ALIVE / STALE / DEAD	Reachability according to the presence policy
Valid, recent RTT	A usable round-trip measurement exists
No interval samples	No measurement in that interval, not 0 ms
Old RTT	Historical evidence; do not display it as current

Accept only direct heartbeat evidence from the correct writer/session.
Reject duplicate, superseded, or invalid acknowledgments.

> **Takeaway:** show freshness alongside the value.

Measurement choices have different costs

Approach	Benefit	Trade-off
Passive heartbeat RTT: current	No extra probe publication schedule	Cadence, loss, and bounded history limit sample availability
DDS protocol counters: current	Detailed repair and queue evidence without a new WAN measurement protocol	Local polling/state; reliable-channel evidence is not every topic's delivery audit
Dedicated probe: alternative	Independent test cadence	Additional endpoints and packets; not the current RTT design

Takeaway: reuse existing traffic first; add probes only for a measured requirement.

Capture now. Classify only after calibration.

```
Per-peer measurements
        -> controlled loss / delay / rate-limit tests
        -> validated thresholds
        -> degradation diagnosis
```

- Separate loss, delay, rate limits, blackout, and host-resource pressure.
- Measure signal delay, noise, and how long a threshold must persist.
- Keep presence-based reachability separate from any future degradation classifier.

> **Takeaway:** protocol counters support diagnosis; they are not a calibrated health score by themselves.

Link-health decision

1. Keep presence, RTT, and delivery pressure as separate facts.
2. Use current heartbeat RTT and per-peer counter baselines.
3. Show freshness and lifecycle changes explicitly.
4. Validate any degradation rule against controlled impairment.

Do not add probe traffic or imply a diagnosis without evidence.

Evidence and further detail

- Current link-metrics design: passive RTT and protocol counters.
- Passive RTT implementation and validation: timing, session identity, freshness, and acceptance gates.
- Presence and health: reachability policy and peer roster.
- Design decisions: capture first; inference requires calibration.

The older dedicated application-ack probe investigation is historical.
This deck describes the current heartbeat-based RTT design.